FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Climate-Smart Mechanized Innovations to Optimize Rice–Wheat Systems in Eastern India

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.2 · Entrepreneurship through Mechanization & Postharvest Services

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

  Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

  Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

  Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

  Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

  Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

  Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

In Eastern Uttar Pradesh (EUP), the predominant rice-wheat cropping system is trapped in a low-productivity cycle. Late planting of rice, followed by delayed wheat sowing beyond the optimal November window, results in significant yield losses (up to 50 kg/ha/day) due to terminal heat stress. This challenge, affecting over 60% of the cultivated area, is exacerbated by traditional, resource-intensive practices like puddled transplanted rice (PTR), which consume excessive labor, water, and energy while contributing to greenhouse gas emissions. This scaling track directly addresses this systemic problem.

The primary goal of this track is to catalyze a systemic shift toward climate-resilient intensification of the rice-wheat system across 21 high-priority districts in Eastern Uttar Pradesh. The core solution is a bundled innovation package centered on **mechanized Direct-Seeded Rice (DSR) followed by Zero-Tillage (ZT) wheat**. This system-level intervention enables a 10-20 day advance in the cropping calendar, unlocking productivity gains and building resilience against climate shocks. The package is further strengthened by integrating DSR-fit, stress-tolerant rice varieties/hybrids, best-bet agronomy, and sustainable service-delivery business models.

This scaling initiative is built on a robust evidence base from years of research and pilots by IRRI, CIMMYT, and partners under projects like CSISA. It has demonstrated yield parity or superiority over traditional methods, alongside significant cost savings and reduced environmental footprint. The track leverages a major bilateral investment—the World Bank-supported UPAGREES project—with IRRI as the technical partner. S4I funding will act as a critical catalyst to deepen technical support, derisk business models for service providers, and accelerate adoption through targeted cluster demonstrations, ultimately moving the system from project-led diffusion to market-led scaling.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **System Optimization Package:** The core innovation of **mechanized Direct-Seeded Rice (DSR)** followed by **Zero-Tillage (ZT) early wheat**.
- **Service-Based Business Models:** Strengthening a network of Custom Hiring Centers (CHCs), Private Service Providers (PSPs), and Farmer Producer Organizations (FPOs) to provide mechanized services on a rental basis to smallholders.
- **Precision Agronomy:** Bundling the core technologies with laser land leveling, integrated weed management (IWM), and best-bet crop management practices.
- **Superior Genetics:** Promotion of short/medium-duration, DSR-fit, and stress-tolerant rice varieties/hybrids, including new herbicide-tolerant (HT) rice options.

2.2 Evidence of Impact

- Mechanized DSR delivers yields comparable to best-managed PTR, while reducing cultivation costs by up to **USD 150/ha** and saving **80% on labor**.
- The DSR + ZT wheat system advances wheat sowing by 10-20 days, boosting subsequent wheat yields by **~19% (approx. 0.5 t/ha)** and raising total system productivity by **~1.0 t/ha**.
- Farmer incomes increase by **USD 100-150/ha** from the bundled package.
- The system reduces GHG emissions by **30-40%** and conserves water by **30-35%** compared to conventional practices.

2.3 Scaling Targets

Smallholder farmers reached	~10,000	> 1 million
Area under DSR + ZT wheat (system)	10,000 ha	> 200,000 ha
Functional CHCs/PSPs strengthened	~50	400+
Value chain actors trained	1,500	10,000 (farmers, operators, extensionists)
New rural enterprises (jobs) created	~500	4,000 - 5,000 (seed, nursery, services)
Investment leveraged	> $5M (from UP-AGREES and state subsidies)	> $70M (from UP-AGREES, ISARC, private sector)

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy

The scaling pathway is a multi-pronged strategy designed to transition from project-led diffusion to a self-sustaining, market-led system. It leverages the significant public investment of the UP-AGREES project while using S4I funds to accelerate and deepen the impact.

- **Public-Platform Integration:** The initiative is embedded within the Government of UP's extension system and the World Bank-supported UP-AGREES project. This ensures alignment with state priorities, access to farmer networks, and integration with government subsidy schemes for mechanization and seeds.
- **Private-Service Economy Creation:** The core pathway to scale is the creation of a vibrant "service economy." S4I funds will be used to de-risk and strengthen the business models of CHCs and FPOs, ensuring they are profitable and sustainable. This involves training operators, establishing local spare part banks, and linking them with financial institutions.
- **Demand Creation through Clusters:** A "cluster-based" approach will establish 10-20 ha demonstration hubs where the full innovation package is implemented with high fidelity. These visible success stories will serve as learning sites to organically attract neighboring farmers and build demand for mechanized services.
- **Strengthening the Enabling Environment:** The track will use evidence from its implementation to advocate for supportive policies, such as cluster-based incentives for DSR adoption. Critically, it will lay the groundwork for integrating carbon credit incentives to provide an additional revenue stream for farmers and service providers, further accelerating adoption.

3.2 Theory of Change

If smallholder farmers in Eastern UP gain reliable access to an affordable, bundled package of DSR-ZT wheat innovations through a strengthened network of local service providers (CHCs, FPOs), **and** critical market failures (lack of access to machinery, quality seeds, and repair services) and coordination failures are addressed through de-risked business models, public-private platforms (Uttar Pradesh Direct Seeded Consortium (UPDSC) / Samriddhi Dhan Network (SDN)), and policy engagement, **then** farmers will adopt the system at scale, leading to **immediate outcomes** of increased yields and reduced costs, **leading to** the **intermediate outcome** of significantly higher farm profitability and climate resilience, **ultimately contributing to** the **long-term, systemic goal** of a transformed, sustainable, and market-driven rice-wheat production system in Eastern India.

3.3 Key Activities Funded by S4I

- **Mobilization and Demonstration:** Expert oversight/contribution and strong technical support in implementation of high-fidelity, large-scale cluster demonstrations to ensure visible success and create demand.
- **Market System Strengthening:** De-risk service provision models by providing deeper technical support and business mentorship to CHCs/PSPs and training a cadre of local mechanics to establish a sustainable repair and maintenance ecosystem (spare part banks).
- **Partnership and Governance:** Support the UPDSC/SDN platform to improve coordination among public and private actors and develop evidence-based policy briefs on carbon credits and DSR incentives.

4. Partners and Roles

Partner Type	Specific Partners and Quantities	Role in Scaling Pathway
Demand Partners	**Farmer Collectives:** ~400 CHCs, 100 FPOs, Women SHGs. **Private Sector:** ITC, Savannah Seeds, Bayer, Corteva, Reliance, Adani Group, SEAL, local machinery dealers.	**Farmer Collectives:** Last-mile delivery of services, seed production, demand aggregation. **Private Sector:** Supply of inputs (seeds, herbicides), off-take agreements for residues, co-investment (CSR), establishing local dealerships for machinery/spares.
Scaling Partners	**Public Agencies:** Dept. of Agriculture (GoUP), UP-AGREES Project, KVKs, State Agricultural Universities (e.g., ANDUAT, BUAT), ICAR-ATARI. **R4D Partners:** IRRI (Lead), CIMMYT, World Bank.	**Public Agencies:** Policy alignment, co-delivery of training, farmer mobilization, institutional ownership, and provision of core subsidy schemes. **R4D Partners:** Technical backstopping, scaling science, MEL, evidence generation for policy, and leveraging additional funding (e.g., ISARC, Bayer DSR Consortium).

Funding Model: S4I catalytic funding will cover the *additional* costs of deepening technical support, derisking business models, and enhancing demonstration quality, while the core costs of machinery subsidies, seed and other input procurements, and large-scale extension are financed by the **UP-AGREES bilateral project and government schemes.**

5. Enabling Environment and Risks

5.1 Opportunities

- **Major Bilateral Investment:** The track is anchored in the $55M+ World Bank-supported UPAGREES project by Govt. of UP, providing a strong financial and institutional foundation.
- **Strong Government Buy-In:** The Government of UP, through the Uttar Pradesh Diversified Agriculture Support Project (UPDASP), Department of Agriculture and the UPDSC/SDN, is a committed partner, eager to scale climate-smart technologies.
- **Private Sector Momentum:** Major agribusinesses like Bayer have committed to scaling DSR in India (to 1M ha), creating a powerful lever for accelerating adoption in EUP.
- **Emerging Carbon Markets:** The growing interest in agricultural carbon credits presents a significant opportunity to create an additional incentive mechanism for farmers to adopt and sustain DSR and ZT practices.

5.2 Challenges and Mitigation

- **Technical Risk (Weed Pressure):** High weed pressure is a major constraint in DSR.
- *Mitigation:* Implement a robust Integrated Weed Management (IWM) program, train service providers and farmers, and introduce new Herbicide-Tolerant (HT) rice varieties as a complementary tool.
- **Institutional Risk (Weak CHC/FPO Capacity):** Newly formed service providers may lack business acumen, leading to financial unsustainability.
- *Mitigation:* Provide a dedicated business incubation component with mentorship in financial planning, operational management, and diversified revenue streams (year-round services).
- **Market Risk (Service Underutilization):** A mismatch between machinery supply and rental demand could lead to asset underutilization.
- *Mitigation:* Use a cluster-based aggregation model to forecast demand before deploying machinery and diversify service offerings. Create local repair networks to minimize downtime and protect operator profitability.
- **Social Risk (Exclusion):** Benefits of mechanization may be captured by larger farmers, excluding women and marginalized groups.
- *Mitigation:* Mandate inclusive targeting in cluster formation (10-30% women farmers, 80-90% small/marginal) and prioritize women and youth for enterprise development and training. Use participatory MEL to track benefit distribution.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Process Outcomes:** Effectiveness of partner coordination (via UPDSC/SDN), fidelity of cluster demonstrations, and reach of training programs.
- **Behavior Change:** Adoption rates of the full DSR-ZT package vs. partial adoption, changes in farmers' and service providers' practices, and demand for new services (e.g., spare parts).
- **System Outcomes:** Verification of adoption rates and area covered (via endline survey early 2027), profitability analysis for both farmers and CHCs, and assessment of the functionality and financial sustainability of the emerging service economy.

- **Inclusion:** Tracking the participation and benefit capture by women, youth, and marginalized groups against the established targets.

6.2 Adaptive Management Learning will be translated into action through a structured adaptive management framework:

- **Quarterly Check-ins:** Regular meetings between the IRRI team and S4I's Area of Work 2 to review progress against milestones and address operational bottlenecks.
- **Annual IPSR Scaling Readiness Reviews:** Participatory assessments using the IPSR methodology will be conducted annually to identify emerging bottlenecks in the enabling environment and refine the scaling strategy.
- **Cross-Project Learning:** Linkages with Area of Work 3 (Enabling Environments) will provide comparative analysis on policy barriers across similar mechanization tracks. A mid-year review (Dec 2026) and annual report (May 2027) will document key lessons.

7. Next Steps and Timeline (2026)

Q1 (Jan-Mar)	• Finalize partner MoUs and operational plans under UP-AGREES/S4I. • Complete baseline diagnostics and suitability mapping for DSR/ZT in 21 target districts. • Initiate training programs for master trainers and mechanics.
Q2 (Apr-Jun)	• Launch first cluster demonstrations for the upcoming Kharif (rice) season. Conduct pre-season hands-on training for farmers and operators. • Strengthen seed systems: initiate decentralized seed production pilots with FPOs/WSHGs. • Facilitate linkages between CHCs and financial institutions for credit. • Establish district-level spare part banks. Implement IPSR/Scaling strategy workshops
Q3 (Jul-Sep)	• Monitor and provide technical backstopping for Kharif DSR demonstrations. • Conduct field days and exposure visits at demonstration sites. • Support private sector partners (e.g., ITC, Bayer) in training their field staff on DSR.
Q4 (Oct-Dec)	• Conduct harvest events, crop cuts, and gather data from the Kharif season. • Mobilize farmers for the Rabi (wheat) season and plan ZT wheat demonstrations. • Complete a mid-year review report, documenting initial adoption rates and lessons learned. •

Communications and Knowledge: Key products will include a **mid-year review report (Dec 2026)**, success stories from farmer-adopters and new entrepreneurs, and policy briefs on carbon incentives. These will be shared through the DSC/SDN platforms, S4I communication channels, and partner meetings.

8. Contact

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